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RF Tagging for Rat Habitats

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Bachelor of Engineering with Honours

Abstract
This project is to create an RFID access door system for lab rats. This device was developed in order to aid the Victoria University of Wellington Psychology Department in their drug and alcohol research, but use can be extended to other rat habitat restrictive research. The device will control access for a designated rat between its social habitat enclosure and the exclusive chamber where it will be able to complete the designated task. The rat's implanted microchip is detected with an RFID antenna and reader module.

Contents

Chapter 1: Introduction

This project is to create a RFID access door system for lab rats. This device is developed in order to aid the Victoria University of Wellington Psychology Department in their research into drugs and alcohol. The device will control access for a designated rat between its social habitat enclosure and the exclusive chamber where it will be able to complete a task. This project will improve both the living conditions of the rats and increase the reliability of the experimental results. This device will be used to aid a variety of different experiments; however, the project was formulated to address the particular scenario outlined below.

1.1. Specific Need Example

The example problem scenario as explained by the client:
In a previous study conducted at Victoria University, rats were observed to see what effect regular alcohol consumption had on the rat's levels of anxiety. Rats are social animals and become lonely if separated from other rats for long periods of time. During the experiments conducted, the rats were forced to be isolated for eight-week periods. They were kept alone in order to ensure accurate alcohol consumption measurements could be taken. Once the eight-week period ended, the rat's level of anxiety was measured. However, the studies were inconclusive. After being isolated for eight weeks, loneliness and depression were more prevalent among the rats than anxiety. The most common test to determine anxiety in rodents is the Elevated Plus Maze [1]. Victoria University of Wellington uses an anxiety test model similar to this. The test is based on rats' natural aversion to open spaces and predators. After subjecting the isolated rats in this experiment to an anxiety test, where the rats were given the chance to stay in a safe nook or explore in wide, bright open spaces, the rats would explore much further than anticipated. While rats are usually scared of unknown humans and bright open spaces, these rats were so depressed and lonely that they would explore further in order to be closer to the human lab technicians, and gain the attention and love that they had been deprived from.

The results of these tests are strongly affected by the fact that the rats had to be kept in solitary isolation. Rats are eminently social creatures, and when they are deprived of company they become depressed [2]. In the alcohol studies, the rats' wellbeing was greater affected by loneliness than alcohol.

1.2. Problem

The problem the client has is accurately measuring the alcohol consumption of each rat while accounting for their social requirements. The current testing scenario creates inaccurate and inconclusive results as well as unnecessarily diminishing the wellbeing of the lab rats involved.

1.3. Aim

The aim of this project is to provide access control between the main area (Zone A) of a rat cage and a restricted experimental zone (Zone B) which only designated rats are able to access. Multiple rats will be contained in Zone A, while certain individuals will have access to Zone B, where they are able to complete a task and return to Zone A. The access control device will be able to grant access to approved rats as well as record the time and date that the rat entered the experimental zone.

1.4. Minimum Viable Product

The minimum viable product was to create a gate device that mechanically separated Zone A and Zone B. The device had to be able to keep a record of which rat had entered through the gate, by means of RFID microchip ID tag (microchip ID). The minimum viable product the difficulty of preventing rats from tailgating one another (a non-approved rat following an approved rat through the gate system). Being able to produce a record of all rats entering Zone B instead of completely restricting access was sufficient for the project.

1.5. Stretch Goal

The stretch goal was to create a higher security system that successfully ensured only designated rats were allowed to enter Zone B. The issue of tailgating rats is addressed.

Chapter 2: Background Survey

This chapter details past research and development in RFID tag use for rodent tracking and rodent access. An animal ethics section includes information on rules and regulation around animal testing.

2.1 Mouse Real-Time Analysis

The paper "Real-time analysis of the behaviour of groups of mice via a depth-sensing camera and machine learning" examines observation techniques that allow for multi-day observation of mice and their behaviour [3]. This paper is relevant to the project because it uses RFID tag implants and RFID detection to improve the accuracy of psychological experiments which measure anxiety over a multi-day period within a normal rodent habitat. Most studies on rodents are limited by the observation window length, as it is difficult to observe animal behaviour constantly for long periods of time. Therefore, it has only been possible thus far to conduct behavioural tests monitoring short term rodent interactions. Unfortunately, there are many cases in which studying rodent behaviour over multi-day periods is desirable. This paper discusses the importance of creating new methods and technology to facilitate multi-day, in-home experiments in the future. When experiments are only conducted in the short term, it is common for the rodents to act differently to how they would if they were accustomed to their testing situation for a multi-day period. The live mouse tracker developed in this paper tracks the location and activity of mice by using RFID sensors (located beneath the flooring of the enclosure), a depth sensing infrared camera and random-forest machine learning. It is able to use these techniques to distinguish mice from the background (irrelevant of their appearance) and observe a mouse's posture, identity and behavioural profile. "We designed an integrated system that tracks and monitors—for hours or days—the activities of four mice in a rich environment. This system determines the outline mask and orientation of each mouse and builds a comprehensive repertoire of individual and social behavioural events from this data" [3]. This study provides advice on working with RFID tags in the 134kHz range (such as the RFID tags that were used in this project), and offers steps to achieve calibration of the RFID reader and code for identifying rodent individuals. This project differs from the live mouse tracker as it also requires a gate system. The main purpose of my device is to address gated access into an area. While there are similarities to this study, my study will use a simpler sensing technique and a detection area sized for rat body measurements instead of mice.

2.2 Instructables RFID Cat Door Tutorial

An RFID cat door tutorial available on Instructables [4], is the most similar device available online to the project. This tutorial details how to engineer a cat flap that can only be opened by the animal wearing a designated RFID tag. The user constructs the physical structure of the cat flap, as well as the electronics and software for their cat door. This cat door is larger than the rat door, but inspired the initial concept used for the project. Fig. 1 shows a photo of the user's completed cat door. The flap design is simple and requires no motors. Animals such as rats and cats are easily trained to open a flap. An important feature shown in Fig. 2 is the locking mechanism of this cat flap that has been achieved by using a solenoid lock.

Fig. 1. Cat Flap with Solenoid Lock [4].

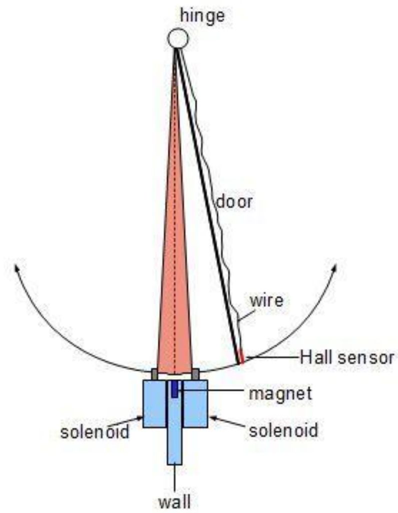

Fig. 2. Solenoid Locking System [4].

2.3 Ethics

To ethically create new technology used with animals in research and commercial use, ethical analysis must be conducted [5] [6]. As engineers, when creating technology that is going to be used on and with animals, it is important to make ethical considerations as to how the technology will be used in order to minimise pain and distress for animals. There is great debate about whether animal testing can be justified, and there are many approaches to justifying and regulating animal testing. The implementation of 'The Three R's'- Replacement, Reduction and Refinement, is commonplace practice in animal testing laboratories, and provides a good starting point for evaluating the engineering of new technology [7].

2.4 ENGR301 Team's Previous Work

In 2019, a Victoria University of Wellington ENGR301 student team worked on this same project [8]. Their final device was not 100% functional, and their design had several problems. The issues reflected on by this team in their Final Project Report provide good insight for the 2022 revision of the project. The main issue that the 2019 team experienced was with the RFID Reader read distance. The RFID reader and antenna used by the team did not have a high enough gain to reliably detect the presence of the passive RFID tags implanted in the rats. They had issues detecting if rat individuals passed through the gate or not, as well as issues with identifying which specific ID tag had been detected. A second issue that this team encountered was incorrect sizing of their gate hole. The team sized their hole too small for an average sized rat to enter through. This meant that below average sized rats had to be used when trialling their device. This was undesirable as most rats were larger than the gate.

2.5 Visit to Animal Psychology Laboratories

The project device is intended to be used within the psychology rat laboratory at Victoria University of Wellington. In order to aid the project design, the laboratory was visited to examine the rats current living conditions and speak to laboratory technicians. Currently the rats are contained in a tower block arrangement, where the cages are stacked in layers of four. Each cage contains two rats each, unless a specific experiment requires this number to differ. Fig. 3 and Fig. 4 below show the current rat cage pods.

Fig. 3. Current Rat Cages Used.

Fig. 4. Current Rat Cages Used

The separate testing room was also visited. In this room the maze device is used for testing anxiety levels in rats. To record the anxiety test experiment, a birds-eye-view camera suspended on a rail system on the ceiling captures the test event. A photo of this configuration is included below in Fig. 5.

Fig. 5. Rat Anxiety Test Maze Structure at Victoria University of Wellington.

Chapter 3: System Overview

3.1 System Overview

The project separates two zones; A and B. Zone A refers to the main habitat zone where rats are housed socially with one or more peers. Zone B refers to the zone compartment where the test substance is available (e.g. alcohol). The zones are separated by Perspex walls, and are connected by a tunnel. The entrance and exit to the tunnel are controlled by door flaps that are locked with solenoid actuator locks. There are three solenoid locks in total. There are two locks on the first door and one lock on the second door. Having two locks on the first door prevents rats from pushing or pulling the door open when locked. The second door only has one lock in order to minimise the chance of a rat becoming accidentally trapped in Zone B. Access control is managed by the detection of RFID microchips. The microchips are detected by a RFID reader and antenna. The system processes the microchip data and sends signals to the appropriate solenoid locks by means of the Arduino Mega 2560 microcontroller.

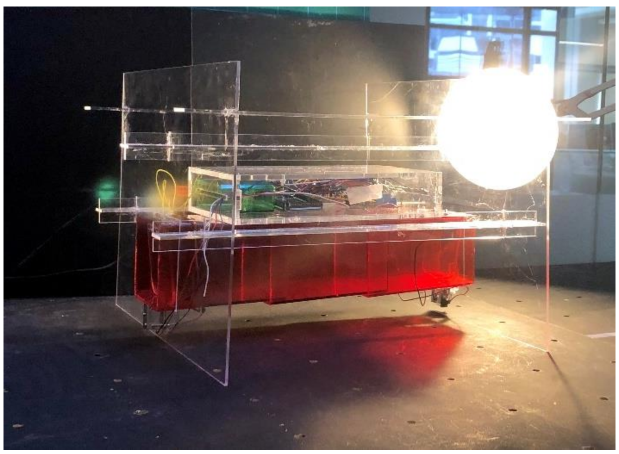

Fig. 6. Photo of the project.

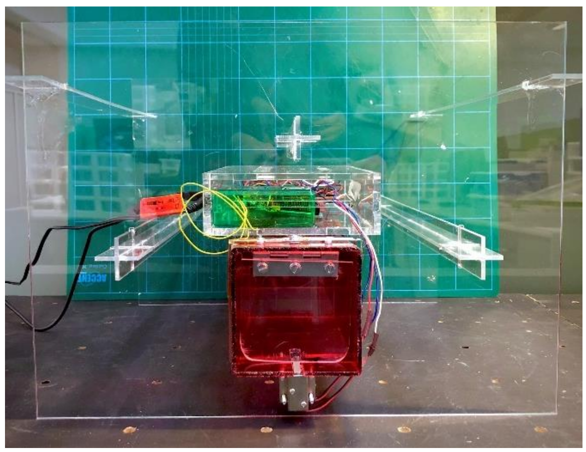

Fig. 7. Front view photo of the project.

3.2 Use Description

The diagrams below give a chronological explanation of how a rat user will use the system. This example shows the use of alcohol as the specific testing substrate.

1. Rat approaches Door 1 . The RFID antenna will scan the microchip in the rat when the rat is within range. If the rat has a microchip that is registered as approved with the system, then locks 1 and 2 will unlock. The rat is now able to push through the flap door and enter the tunnel.

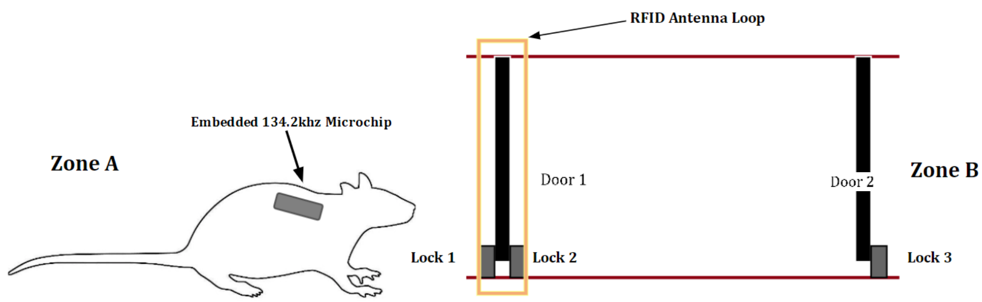

Fig. 8. Rat Use Description 1.

2. There is a 10 second wait time to allow the rat to enter the tunnel and approach Door 2. After the 10 seconds is up, Lock 2 locks and Lock 3 unlocks. Lock 2 locking prevents other rats from entering the tunnel. Lock 1 stays unlocked at this point so that if the rat wishes to exit suddenly, it is still able to.

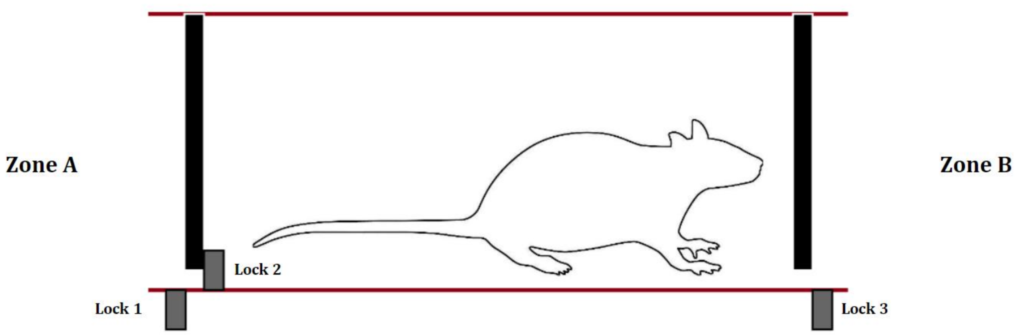

Fig. 9. Rat Use Description 2.

3. Now that Lock 3 is unlocked, the rat can push through Door 2 to Zone B. The rat is able to drink from the alcohol sample. After another 10 second wait time, all locks lock.

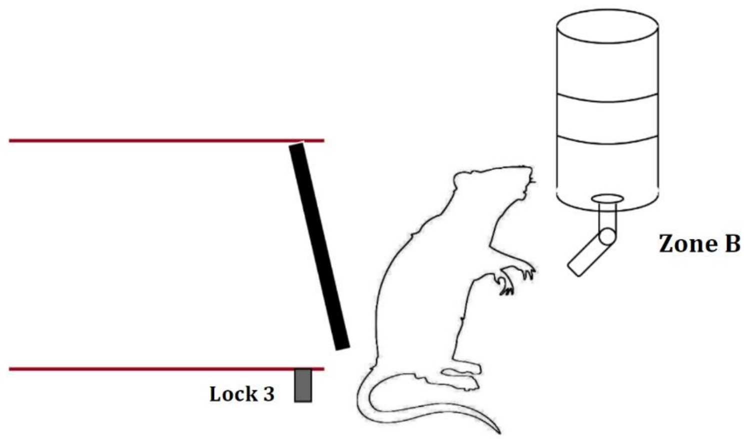

Fig. 10. Rat Use Description 3.

4. Door 2 is only locked in one direction. Any rat in Zone B is always able to exit back into the tunnel. Once the rat has finished drinking the sample, it can push through Door 2 and re-enter the tunnel. Once in the tunnel it will be detected by the RFID antenna. This will cause the lock

routine to run again. Locks 1 and 2 will therefore be unlocked for 10 seconds and the rat will be able to push through Door 2 and exit.

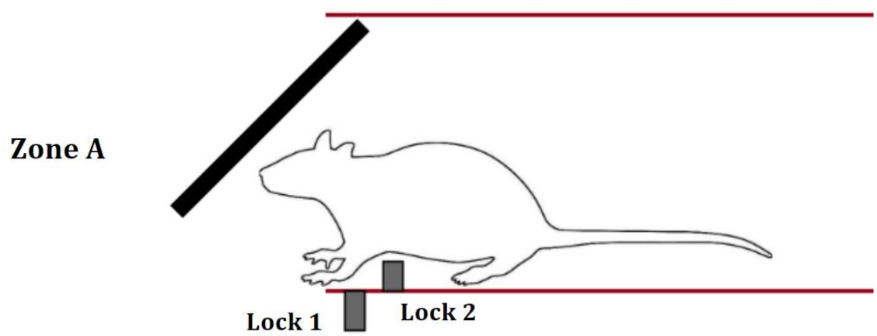

Fig. 11. Rat Use Description 4.

3.3 Physical Structure

As rats often chew items available to them in their habitat, it was important to design a structure that was made out of material that is non-toxic to rats when ingested [9]. The material must be hard enough to withstand chewing attempts during long term exposure to rats. Clear, brittle plastics are most commonly used in rat enclosures.

The walls, support beams and wire storage box components of the project were laser cut out of 3mm thick clear Plexiglass acrylic sheets.

The two walls with a tunnel between them imitate the airlock structure used in pest control efforts in nature reserves [10].This airlock structure is implemented to mitigate the possibility of non-approved rats tailgating approved rats for access into Zone B.

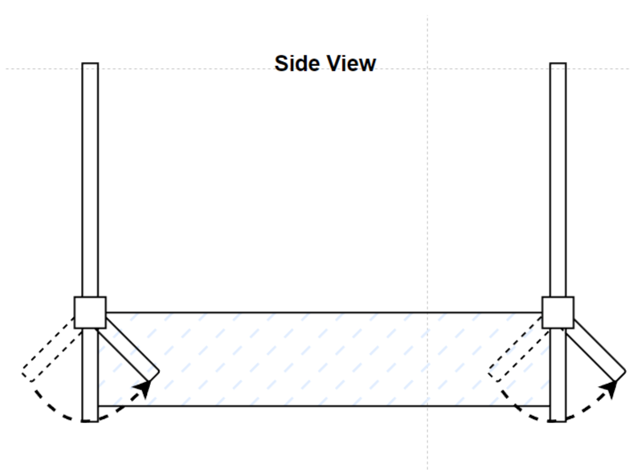

Fig. 12. Side view diagram.

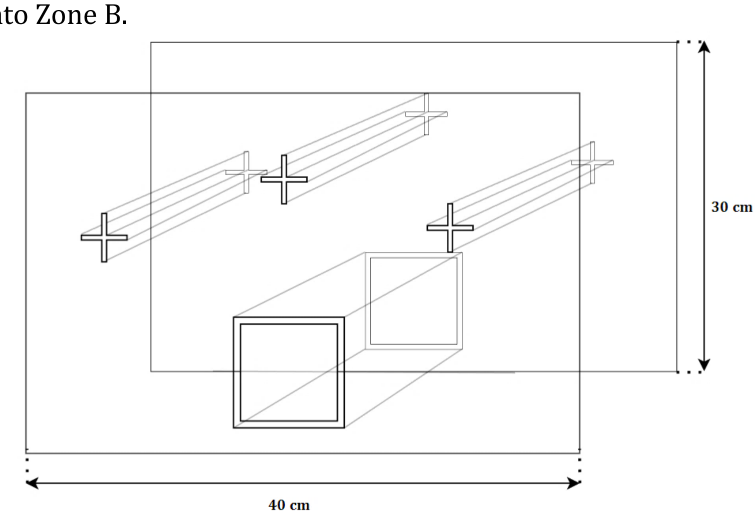

Fig. 13. Front view diagram.

Fig. 14. Photo side view of project.

The circuit boards, Arduino Mega 2560, and RFID reader are stored within a box between the walls. The box keeps the wires and hazardous components out of reach from the rats. This is shown in Fig. 15.

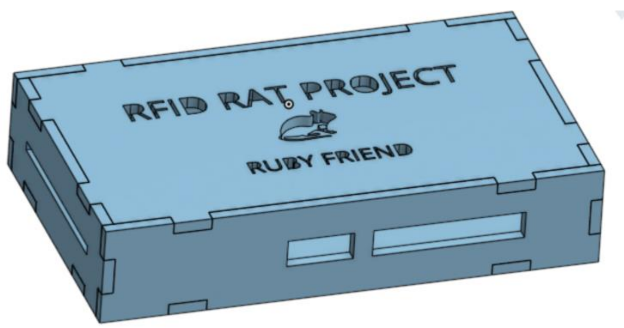

Fig. 15. CAD model of wire storage box.

3.4 Rat Use Case Diagram

The flow diagram in Fig. 15 below shows the function of the system. The red loop represents the action of the rat. The grey loop shows the order in which the functions of the project occur. The dotted lines show at which points in time the rat's action matches the stage of the system.

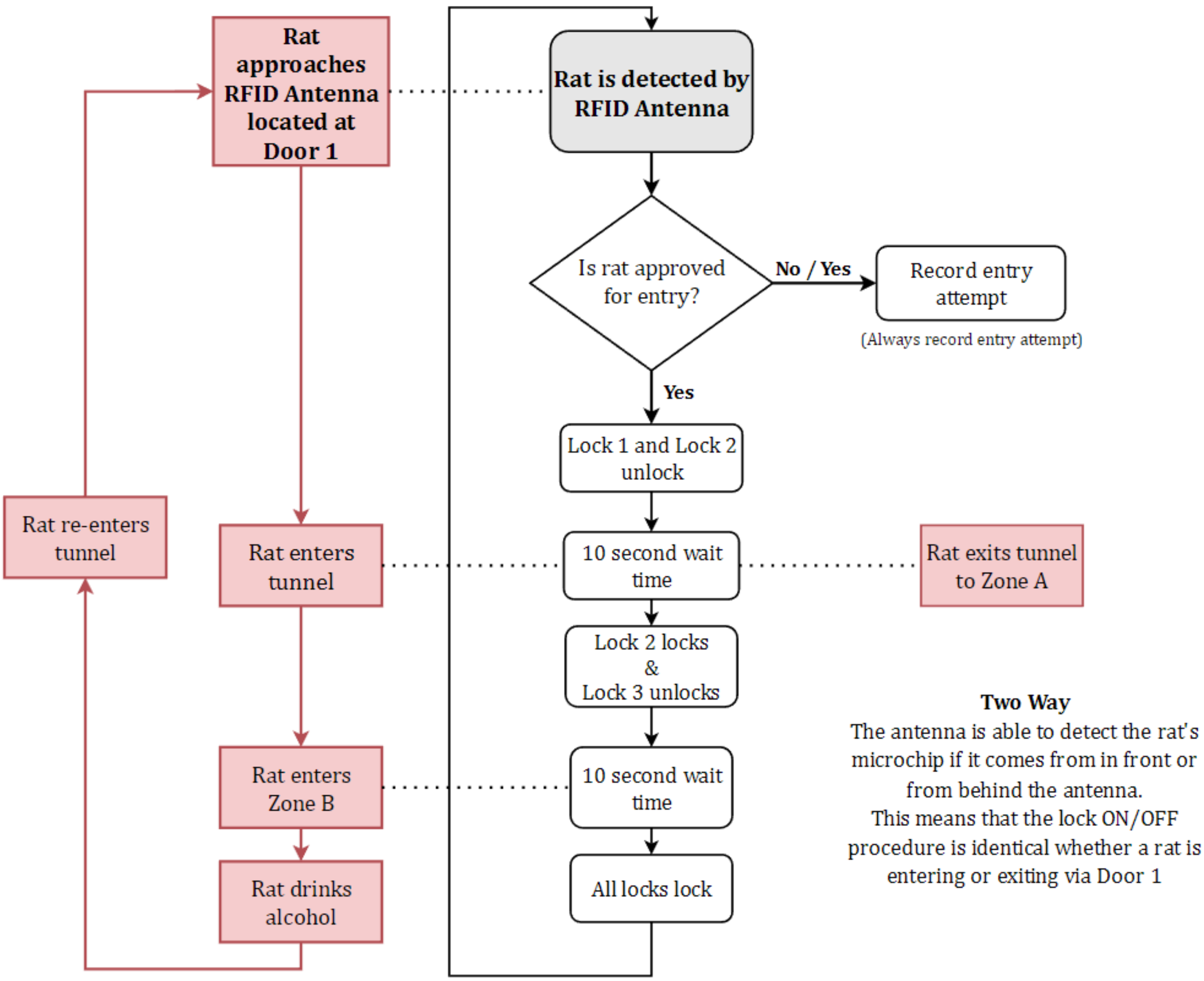

Fig. 16. Use case flow diagram.

3.5 Electronics

The main electronic components used in the project are as follows:

- Arduino Mega 2560
- Four Channel 12V Relay
- 3x 12V Solenoid Actuator Locks
- RFID Reader
- RFID Antenna

- RFID Transponder Microchip

The antenna picks up data from available microchips. This is processed by the RFID reader and sent as a hex format number to the Arduino. The Arduino then sends the appropriate ON and OFF signals to the necessary solenoids.

The Arduino provides a 5V output. This is not enough to power the RFID reader or solenoid locks which each need 9-12V to run. A four-channel relay module was chosen to connect the Arduino to the other components.

The wiring diagram in Fig. 18 shows how the components of the circuit are connected. The output pins PIN4, PIN5 and PIN6 are connected onto INPUT 1, INPUT 2 and INPUT 3 of the relay module. The output channels CH2, CH3 and CH4 then send ON/OFF signals to the required Lock 1, 2 or 3. The RFID reader has four output lines. Only three of them are used in the project. The RFID reader transmit input is connected to PIN10 of the Arduino. This is how the RFID reader passes its data to the Arduino. The input voltage is connected to CH1 of the relay. The ground line for all components connects to a common ground.

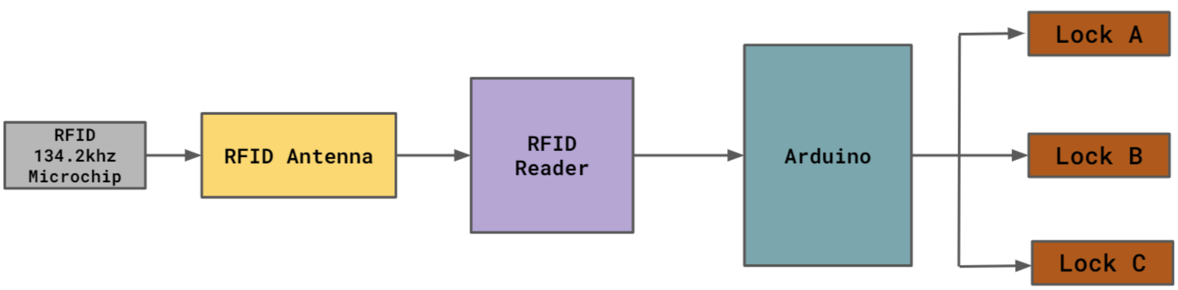

Fig. 17. Electronics component block diagram.

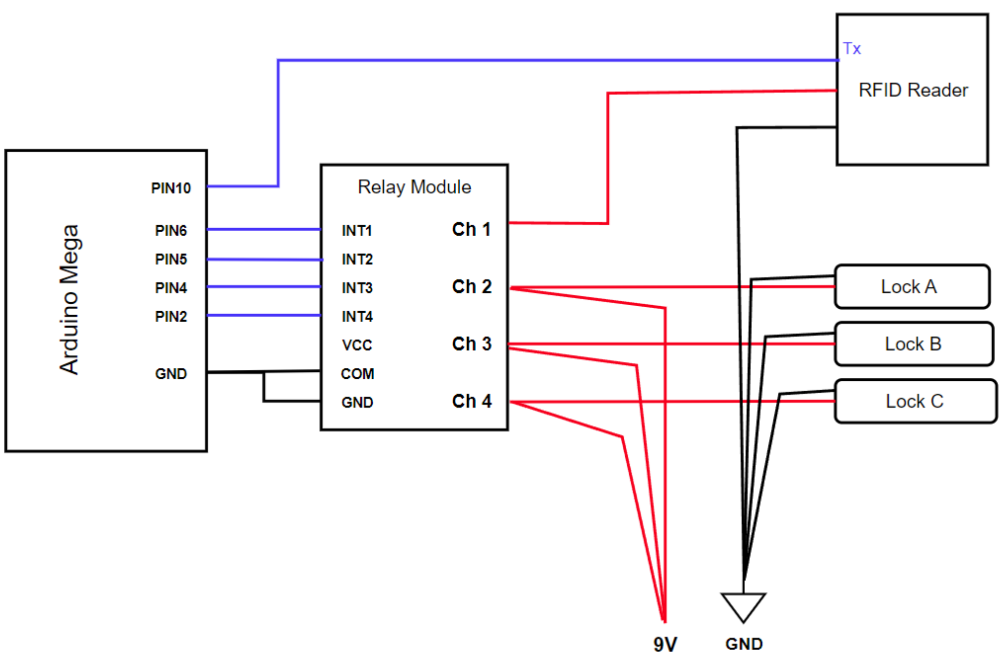

Fig. 18. Wiring diagram.

Chapter 4: Feature Design and implementation

4.1 Safety vs security

The majority of design constraints and considerations are matters of safety vs security. As we are working with animals, animal safety and wellbeing is of utmost importance. The goal of the project is to create a high security gate system that prevents unwanted rats from reaching certain areas. However, when designing most features of the project, an increase in safety minimises security. Each design feature must balance safety and security.

4.2 Door

The initial plan for the lock and door was to attach a lock onto the door as demonstrated in the RFID Cat Door tutorial (Fig. 2). However, this raised the issues of wires exposed for rats to chew and a door that could be too heavy for a rat to push through. Having a lock on the ground that rests against the door is better because this will stop the door from becoming too heavy for the rats to push. If the rats are unable to push the doors open, then the project would be unsuccessful. Having wires attached to the door and exposed also raised issues. In the final design, wires are able to be hidden under the ground.

The gap around the door took a surprising amount of fine tuning. This was a prime example of the safety vs security trade-off. When initially considering an appropriate tunnel length, the plan was to make sure the tunnel was long enough for a rat's whole body and tail. The common subspecies of rats used in laboratories is the *Rattus norvegicus domestica;* with an average body length of 25 cm a tail length of approximately 15cm [11]. Having a tunnel shorter than 40 cm (approximate rat tail plus body length) is more practical, however by shortening the tunnel, if the rats' body and tail do not fit in the tunnel, then the risk of injury to the rat is heightened (by trapping its tail or body in the door) and security of the system (if the door is unable to shut then other rats can easily tailgate) is compromised. To deal with the tail entrapment issue, a gap was added underneath the door. This gap will allow the rat's tail to poke out, so that the tail is safe from harm and does not prevent the door from shutting. This design feature aids security and safety. The gap must not be too large that the whole rat's body is able to slip through the hole. This door and gap design are shown in Fig. 19.

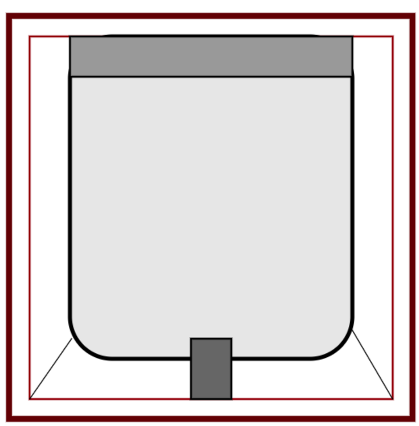

Fig. 19. Diagram of doorway. [Note: the gap shown around the door flap].

4.3 Tunnel

The tunnel component of the project is 30 cm long red Perspex. Due to having no red visual cone cells in rat's retina, rats are unable to see the colour red [12]. Due to this fact, it is common for rat enclosures to contain red-tinted tunnels and huts; the lower visibility providing the rats a sense of sanctuary, and the transparency allowing for lab technicians to see through the red coloured walls and observe the rat's behaviour within [13].

Fig. 20. Side view diagram of tunnel.

4.4 Lock

A configuration of three solenoid actuators was used in the project. The body of the lock must be short enough to fit in the space between the tunnel exterior and the ground. The lock's latch needed to be long enough to protrude into the tunnel and block the swing of the door.

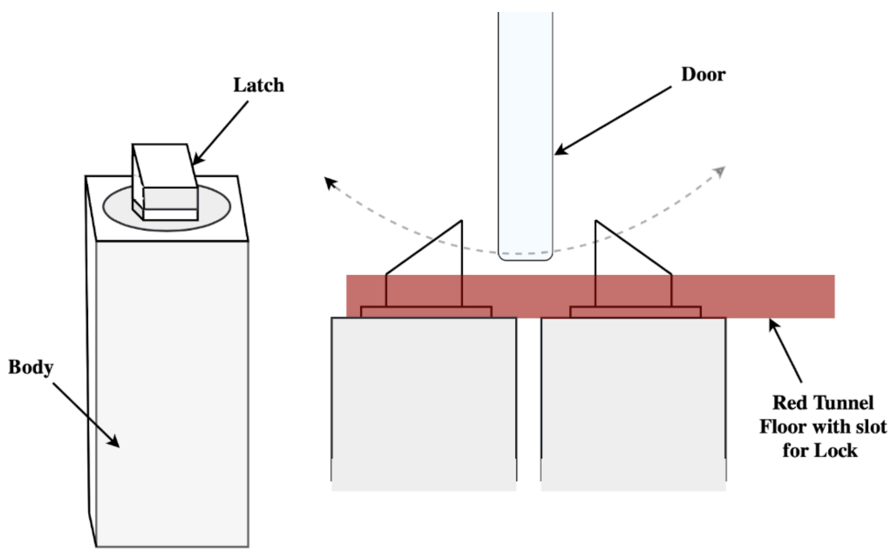

Fig. 21. Lock diagram.

When the door is locked the solenoid is in the raised OFF position. When the solenoid is powered ON, the latch retracts into the locked body and unlocks the door. When the actuators are in the OFF state, pressure from a closing flap door is able to push down the lock. It bounces back to a fully extended

OFF position when no pressure is applied. The lock has a tapered triangle latch. This allows the door to click back into the locked position, even if the door is held open after the latch of the lock raises.

The activation voltage and current requirements of the solenoids are higher than the Arduino is able to provide. A relay needed to be implemented into the circuit to receive control signals from the Arduino and then supply the voltage to the solenoid coils.

4.5 RFID

When considering an RFID reader component, the first decision to make was whether to buy a pre-configured RFID reader component or to construct the circuit from scratch. It was decided that a pre-made circuit made the most sense. The intention of the project was not to construct an RFID reader, and doing so would be an unnecessary time engagement.

A functional RFID reader model requires a reader, antenna and a transponder (microchip). The reader sends a signal to the antenna which transmits electromagnetic radio waves from the coil to scan for any nearby microchips.

There is the option of passive or active RFID tags. This project uses passive RFID tags which hold no power source of their own, and are only activated when power is transmitted to them from the antenna. This activates their circuit, and the tag can transmit back the read-only coded message. Passive RFID tags are more suitable for animal use as they are inexpensive and require neither charging nor maintenance.

A 134.2kHz FDX-B ISO11784 reader module was found and purchased. The only place that sold this was Aliexpress.com. This is not a desirable first choice of retailer as it has a history of being unreliable for electronics. Fortunately, the unit arrived within two weeks of ordering it and functioned as promised.

The read distance of the reader was the initially the most pressing design concern. Commercial readers used for scanning pet microchips in veterinary clinics are expensive and have a very limited read distance [14]. They need to be pressed flat on the animal's nape to get a reading. This project must read the rat at a distance of 5cm.

[Note: It is important to note that the antenna output is 360 Volts peak to peak. This level of voltage will cause injury if the circuit is touched while it is running. The RFID reader for this project is kept in a separate enclosure to prevent handling.]

4.6 Transponder

Concerns of whether the desired read distance of 5 cm could be achieved, prompted considerations of the type of which transponder type would be the most appropriate.

An active RFID tag has the benefits of a longer range due to having its own source of transmission power. However, these tags must be attached to an animal externally by means of a harness or collar. Rats are gnawing animals with incisors that continuously grow [9]. In order to maintain their dental health, they must regularly chew to wear down their teeth. In conjunction with being curious animals, it concludes that minimal material must be exposed to rats, and this must be nontoxic. This leads to the decision for picking a medically inserted passive microchip transponder.

Animal RFID reader and transponder modules are available in two common frequency bands; 125-kHz and 134.2-kHz [15]. These frequencies are referred to as "low frequencies". The low frequency means the signal is able to penetrate water, air and flesh easily; meaning that when implanted in an animal, the tag can still be detected. However, because they are low frequency they are only capable of transmitting a small amount of data. For our purpose, each rat individual's identification number is the only thing that must be stored on the chip. Different antenna configurations are needed for different frequencies of microchips. Many countries have adopted the ISO regulation of 134.2-kHz being the standard for pet microchips. The Victoria University of Wellington already owns 134.2-kHz microchipping instruments. While they were comfortable with the project exploring other options, it has been discovered that 134.2-kHz is coequally adequate to a 125-kHz option.

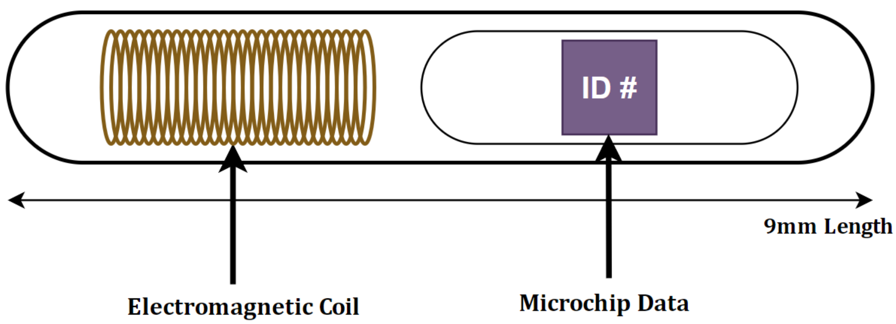

Fig. 22. Diagram of microchip transponder.

4.7 Antenna

The antenna used is the one that arrived as part of the RFID reader module set. Originally it was intended that the antenna for this project would be constructed out of a coil of electromagnetic wire. However, when the RFID reader arrived, once tested, the antenna proved to be ideal for the detection of the microchips. It was not necessary to construct an additional antenna as the one provided was accurate.

The type of antenna used in this project is a 100mm x 100mm square planar spiral coil. It is an air core conductor; meaning that there is no ferromagnetic core in the centre of the coil [16]. Air cores have the advantage of the inductance being unaffected by the amount of current carried in the coil. However, the downside is that a higher gain or higher number of turns is required in order to achieve an inductance value equal to a coil with a ferromagnetic core.

RFID readers receive their transmit power from the RFID reader [17]. The antenna transmits a signal to nearby RFID microchip transponders by electromagnetic wave propagation. If a microchip is in the path of the RFID antenna's magnetic field then it has a chance of being detected by the RFID reader. The signal strength and field coverage of the antenna is partly determined by the wave's expansion as it leaves the antenna. A wave with further degrees of expansion will have a bigger field coverage, yet less signal strength.

4.8 Code

When the device is connected to power, the Arduino will implement the software required to run the system. If the RFID reader is connected through the Arduino pin and available, the RFID reader will begin scanning for nearby microchips. If a microchip is detected, the data will be sent from the reader to the Arduino. The number received is a hex format ASCII number. This number is converted into a 15-digit decimal number. The date, time, ID number and approval status is saved to a text document record. Once the ID number is found, if the ID number matches an approved ID number registered in the system, then access will be approved. The solenoid locks will be subsequently sent the correct ON and OFF signals. Fig. 25 shows a flow chart of the code stored on the Arduino.

These chips use the FDX-B protocol [18]. This is a common format used in animal identification transponders. It is described fully in the ISO 11784 & 11785 publications [19]. This protocol for storing information is shared between manufacturers and therefore is a standardised ID number format. The first three digits of the number refer to the country code. This is followed by a one-digit spacer, a three-digit manufacturer's number and then the eight-digit unique ID. An example number is shown in Fig. 24.

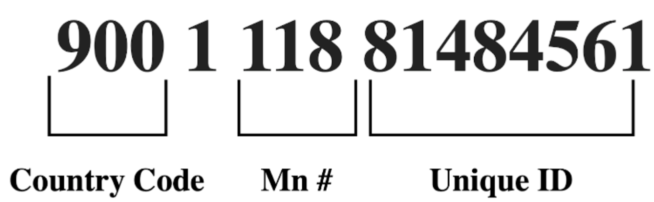

Fig. 23. Microchip ID number example.

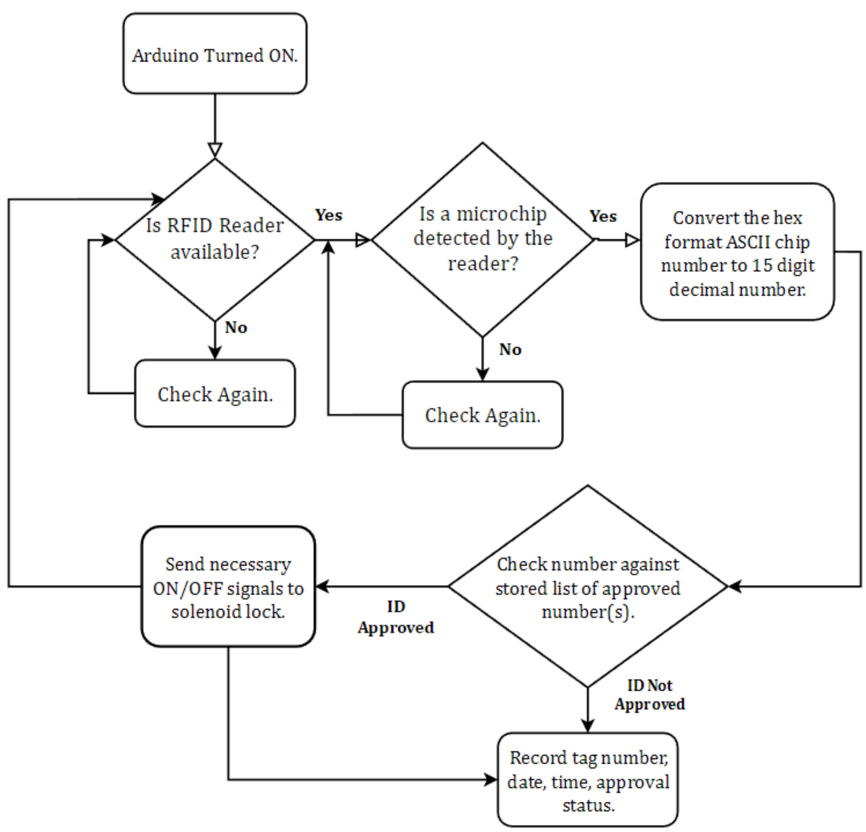

Fig. 24. Software flow diagram.

Chapter 5: Results

5.1 RFID distance testing

To test that the RFID reader would be able to pick up a rat's microchip successfully, multiple tests were conducted to test the range of the reader. The ideal microchip detection distance is 5 cm. This distance is ideal because the distance between the lip of the tunnel and the antenna coil is 4 cm. Therefore, a distance of 5 cm would ensure that rats which intended to enter the tunnel would be registered and approved before they attempt to push the door open.

Once accurately configured, the reader and antenna were tested to see what range was accomplishable. Tests were conducted to find the maximum read distance with different conditions. The microchip transponder had different maximum read distance when held parallel to the antenna in contrast to when positioned perpendicular to the antenna. The transponder was also trialled, clasped tightly in a hand, so that the direct path for RFID was blocked by flesh. The width of the hand was approximately 2cm. This is a thicker depth of flesh than the microchip will be subjected to when imbedded in a rat and therefore surpasses the necessary trials for making sure a flesh boundary does not negatively affect the microchip readability. However, the submersion in flesh has less of an impact than expected, and whether the transponder was perpendicular or parallel to the reader had a much larger effect on the read distance.

The test results are available in the table below. The table simply states whether a distance was achievable and under which conditions.

Max. Read Distance (mm)		Orientation: Perpendicular	Orientation: Parallel
Substance	Flesh	124	52
Substance	Air	131	74

Fig. 25. Microchip read range results table.

The project relies on accurate detection of microchips. To perform the maximum read distance testing, the microchip was held and slowly moved closer to the antenna. Tests were conducted to analyse the effects of the microchip orientation as well as to analyse the effect of a flesh barrier between the microchip and the antenna. When the microchips are injected in the rat’s shoulder, they will be submerged approximately 4 mm in flesh. It would prove too difficult, inaccurate and unnecessary to perform these trials with a live microchipped rat. So, to test the maximum read distance when submerged in flesh, the microchip was clasped inside a human fist. This meant that the microchip was effectively submerged in 15mm of flesh. Each trial was completed five times and an average value was taken, which is shown in Fid. 26.

The testing indicated that the orientation of the microchip with respect to the orientation of the antenna plane had a significant influence on the read range capability. The range of the reader varied between 5.2cm and 12.4cm, depending on whether the microchip was held parallel or perpendicular to the reader. The maximum distance for each case was achieved when the microchip was held perpendicular to the plane of the antenna. The range was greatly reduced when the microchip was held parallel to the antenna.

Once the microchip is implanted into a rat, it is common for it to move around. It cannot be guaranteed that the microchip will sit in the rat in a desired orientation. Therefore, there will be a range of possible readability. A research project detecting RFID microchipped wild rats at Melbourne University showed similar results [20].

5.2 Rat comfortability Testing

Rats were introduced to the device and left to supervised play. The rats were observed and evaluated for three things:

1. **Safety**
    - Do the rats gnaw on, or express interest in gnawing on exposed elements of the project?
    - Do the rat's tails, heads or limbs become stuck in any mechanical elements?
2. **Usability**
    - Is the use of the tunnel natural to the rats?
    - Do the rats instinctively know how to push through a cat flap style door?
    - Do the rats pull on the doors as well as push?
    - Do the rats attempt to tailgate their peers?
3. **Comfortability**
    - Do the rats express behaviours indicating stress or discomfort at any point?

The trials consisted of two, separate one-hour slots of rat play time. Due to time constraints, a full eight week testing period that imitates the alcohol testing trial length (outlined in section 1.1) was not be feasible.

Findings

During the two-hour long sessions, several behaviours exhibited by the rats were relevant to the project.

1. During the supervised play sessions, no rats were injured or came close to injury.
    - The rats did not attempt to gnaw or chew on anything.
    - No rat extremities were trapped in mechanical components.

2. The rats immediately figured out the intended use of the push flap doors and tunnel.

- Interestingly, the rats showed immediate intelligence when presented with a one-way locked push door. The rats were instinctively capable of reaching their hand into the gap to pull the door towards them instead of pushing.
- In the sessions no rats attempted to tailgate another rat.

3. The tunnel was a good height above the ground for them to step into comfortably.
    - The rats could fit comfortably in the tunnel, and even exhibited their ability to turn around when inside the tunnel. Fig. 27 below shows a photo captured of a rat test subject half way through turning direction inside the tunnel.

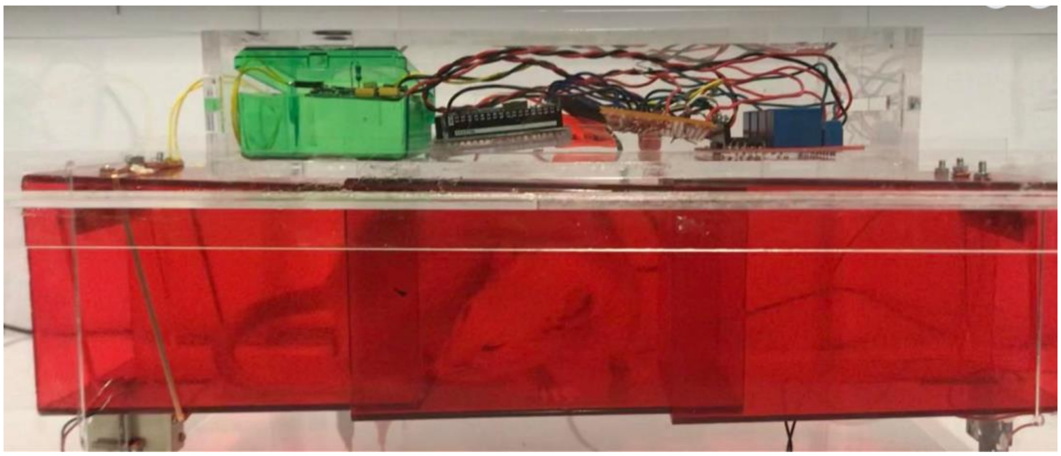

Fig. 26. Rat demonstrating ability to turn around in tunnel.

- Often the rats would spend time in the tunnel for leisure as well as just a means of travel. This meant that they would often spend longer than the estimated 10 seconds inside the tunnel. While this is positive and indicates that the rats feel comfortable when in the tunnel structure, it means that the rats may use the tunnel for play instead of the intended travel purposes. Fine tuning of the lock delay sequence times is needed.
- Interestingly, another measure of rats' comfortability can be observed by their grooming behaviours. Rats regularly groom themselves, however if a rat is in an area in which they feel threatened or unsafe, they will not groom themselves[592], instead waiting until they find an area in which they deem safe before they start grooming. During the observation session, a rat was observed grooming itself while sitting in the tunnel. An image of this is shown in Fig. 28 below.

Fig. 27. Photo of a rat grooming itself from within tunnel.

Chapter 6: Evaluation

6.1 Future Work

6.1.1 More Testing Needed

The initial goal of creating a project to fulfil the needs of the specific eight-week rat alcohol anxiety project has been addressed. However, due to time constraints the eight-week testing period has not yet been trialled with the project. Testing the project for the eight-week period is an important next step. This will highlight improvements and successes presented by this version of the project.

6.1.2 Refining Access Routine

It is expected that after a thorough testing period, fine tuning will be needed in the lock sequence, especially regarding the delay timings. Currently the linear process of 10 second delays and locks opening and closing fulfils the purpose. However, creating a more secure system where rats do not have to wait exceedingly long time intervals to progress to the next stage will make for a smoother testing environment. It is expected that revision of the timing intervals will be the first necessary step in improving system, however there are also several further steps that would make for a more robust system, outlined below.

Currently the RFID read lock sequence is identical whether the rat is entering or leaving Door 1. This system works and is simple. A software format that differentiates between a rat entering and exiting could solve security issues. However, if a rat remains in the tunnel section for longer than expected, the chip could be detected twice and initiate undesired repeating lock sequences. This could cause a rat to be trapped. As an issue of safety, the current multi-use sequence removes the chances of a rat getting stuck; as every time an approved microchip is scanned, Door 1 will unlock. However, if a rat with a non-approved microchip manages to tailgate into the tunnel, and then remains there until all locks relock, its non-approved microchip ID will not allow Door 1 to open. This is an issue. Below are some technology advancements that could address this issue in the future.

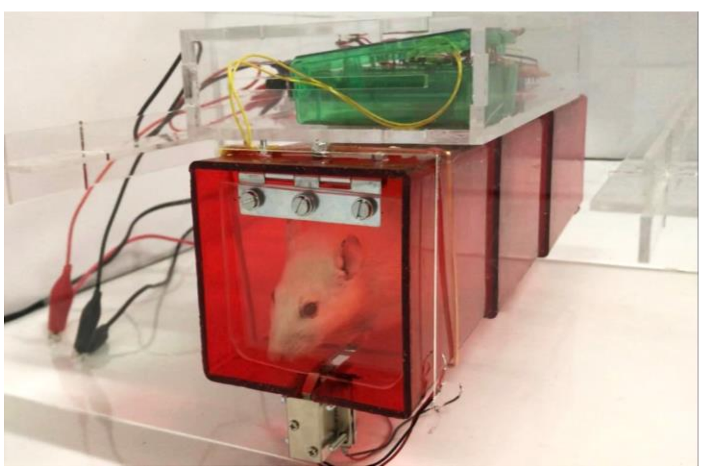

Fig. 28. Photo of non-approved rat pushing on door with lock locked.

Adding a sensor inside the tunnel would allow the device to regularly monitor the location of the rat and regulate the status of the locks accordingly. By using an infrared or camera style sensor, the device would be able to check if any rats are trapped inside the tunnel. As well as solving this safety issue, the sensor would aid in data collection, observing how rats interact with the structure.

Another approach to this issue is to add a second RFID reader to Door 2. If each door's lock sequence was separate and each were separately activated by scanning the microchip, then this would solve security issues. The downside is that this will increase the production cost of the project. Also, RFID antennas such as the one used in this project do scan two ways. Shielding would need to be trialled and implemented to make sure that each reader only scans in the desired direction. This could be done with an RFID shielding material.

6.1.3 User Interface for Victoria University Psychology Department

Currently the software works. If the USB of the Arduino is plugged into a computer then the RFID information and status is printable onto the Arduino serial monitor. When not plugged into a computer, the project still runs the same, but no data is viewable. Currently if the lab technicians wish to update the list of approved rat access IDs they will have to have knowledge of C++ coding. Creating a user interface module would make use of the project more accessible for people without coding knowledge, and just simpler in general.

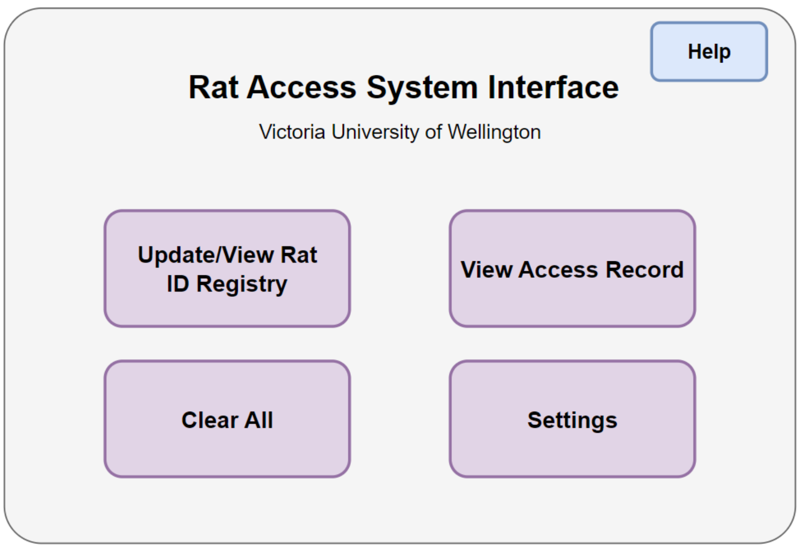

Fig. 29. Example diagram of possible User Interface homepage.

Chapter 7: Conclusion

The project achieves the following goals:

- A record is kept of the rat entry and exits, including time, date, ID number and approval status.
- The project prioritises the safety of rat participants.
- A RFID module solution is implemented to successfully detect a rat at the desired distance of 5cm.
- A solution was implemented to prevent rats from tailgating one another.

The project achieves these goals with an airlock style two gate system, where Door 2 will not unlock until Door 1 is locked. This is the main premise which ensures a secure access system. Access is monitored with an RFID module system. From initial testing, a maximum read range of the microchip by the RFID antenna was found to be approximately 5.2 cm - 12.4 cm. When rats were tested with the project in the sessions, it was observed that not only was the device safe, the rats felt comfortable using it. The rats were able to figure out how to push through the flap style doors straight away, and it was no hindrance to their exploration. Future work should include further fine tuning of the lock sequence routine via extensive testing and the addition of a sensor to the inside of the tunnel.

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